



Yakeen Plus (2025)

Practice Test - 09

DURATION : 200 Minutes

DATE : 20/11/2024

M. MARKS : 720

Topics Covered

Physics :	Dual Nature of Radiation and Matter, Atoms, Nuclei, Semiconductor Electronics: Materials, Devices and Simple Circuits
Chemistry :	The p-Block Elements (Group 13,14,15, 16, 17, 18), Salt Analysis, Practical Physical Chemistry
Biology :	(Botany): Organisms and Population, Ecosystem, Biodiversity and Conservation (Zoology): Biotechnology and its Applications Evolution Biotechnology: Principles & Processes

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours 20 min.** duration.
3. The test booklet consists of **200** questions. The maximum marks are **720**.
4. There are four Section in the Question Paper, Section I, II, III & IV consisting of Section-I (**Physics**), Section-II (**Chemistry**), Section-III (**Botany**) & Section IV (**Zoology**) and having **50 Questions** in each Subject and each subject is divided in two Section, **Section A** consisting 35 questions (all questions are compulsory) and **Section B** consisting 15 Questions (Any 10 questions are compulsory).
5. There is only one correct response for each question.
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number : _____

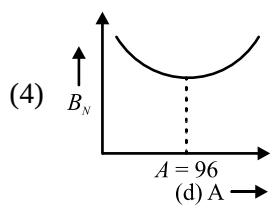
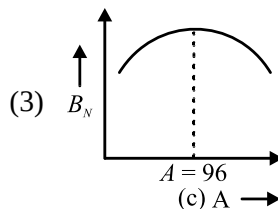
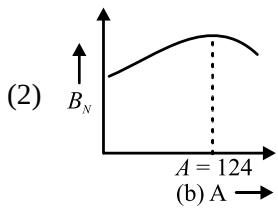
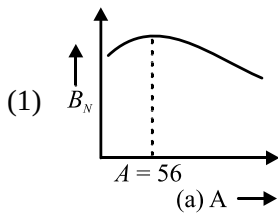
OMR Bar Code Number : _____

Candidate's Signature : _____ **Invigilator's Signature** _____

SECTION-(I) PHYSICS

SECTION - A

1. The dependence of binding energy per nucleon, B_N , on the mass number A , is represented by:



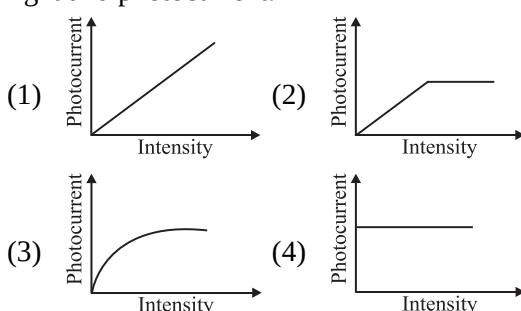
2. Photoelectric emission occurs only when the incident light has more than a certain minimum?

- (1) Frequency (2) Power
(3) Wavelength (4) Intensity

3. The threshold frequency for a photosensitive metal is 3.3×10^{14} Hz. If light of frequency 8.2×10^{14} Hz is incident on this metal, the cut-off voltage for the photoelectric emission is nearly:

- (1) 5 V (2) 1 V
(3) 2 V (4) 3 V

4. Choose the **correct** graph of intensity of incident light and photocurrent.



5. The velocity of most energetic electron emitted from a metallic surface is tripled when the frequency ν of the incident radiation is doubled. Then work function of this metal will be (h is Planck's constant):

- (1) $\frac{5h\nu}{8}$ (2) $\frac{3h\nu}{8}$
(3) $\frac{7h\nu}{8}$ (4) $\frac{h\nu}{4}$

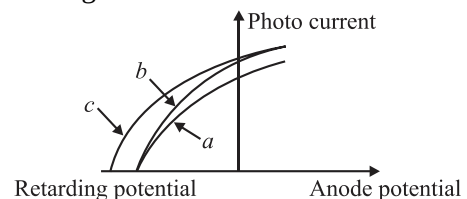
6. The work function of metal is 1 eV. Light of wavelength $3000 \text{ \AA}$ is incident on this metal surface. The maximum velocity of emitted photoelectron will be:

- (1) 10 ms^{-1}
(2) $1 \times 10^3 \text{ ms}^{-1}$
(3) $1 \times 10^4 \text{ ms}^{-1}$
(4) $1 \times 10^6 \text{ ms}^{-1}$

7. If m is the mass of an electron and c is the speed of light the ratio of the wavelengths of photon of energy E to that of the electron of the same energy:

- (1) $c\sqrt{\frac{2m}{E}}$ (2) $\sqrt{\frac{2m}{E}}$
(3) $\sqrt{\frac{2m}{cE}}$ (4) $\sqrt{\frac{m}{E}}$

8. The figure shows a plot of photo current versus anode potential for a photo sensitive surface for three different radiations. Which one of the following is a **correct** statements?



- (1) Curves (a) and (b) represent incident radiations of same frequency but of different intensity.
(2) Curves (b) and (c) represent incident radiations of different frequencies but of different intensity.
(3) Curves (b) and (c) represent incident radiations of same frequency having same intensity.
(4) Curves (a) and (b) represent incident radiations of different frequencies and different intensities.

9. In any fission process the ratio $\frac{\text{mass of fission products}}{\text{mass of parent nucleus}}$ is:
- Less than 1
 - Greater than 1
 - Equal to 1
 - Depends on the mass of parent nucleus
10. Which of the following are suitable for the fusion process?
- Light nuclei
 - Heavy nuclei
 - Element must be lying in the middle of the periodic table.
 - Middle elements, which are lying on binding energy curve.
11. Masses of nucleus, neutron and protons are M , m_n and m_p respectively. If nucleus has been divided into neutrons and protons, then:
- $M = (A - Z) m_n + Z m_p$
 - $M = Z m_n + (A - Z) m_p$
 - $M < (A - Z) m_n + Z m_p$
 - $M > (A - Z) m_n + Z m_p$
12. If M_0 is the mass of an oxygen isotope ${}^{17}_8\text{O}$, M_p and M_N are the masses of a proton and a neutron respectively, the nuclear binding energy of the isotope is:
- $(M_0 - 8M_p) C^2$
 - $(8M_p + 9M_N - M_0) C^2$
 - $M_0 C^2$
 - $(M_0 - 17 M_N) C^2$
13. Fission of nuclei is possible because the binding energy per nucleon in them:
- decreases with mass number at low mass numbers.
 - increases with mass number at low mass number.
 - decreases with mass number at high mass number.
 - increases with mass number at high mass number.
14. When aluminium is bombarded with fast moving neutron, it changes into sodium with emission of particle X according to equation?
- $${}^{27}_{13}\text{Al} + {}^1_0\text{n} \longrightarrow {}^{24}_{11}\text{Na} + \text{X}$$
- The particle X is:
- Electron
 - Proton
 - Alpha particle
 - Positron
15. The binding energies per nucleon for a deuteron and an α particle are x_1 and x_2 respectively. What will be the energy Q released in the reaction?
- $${}_1\text{H}^2 + {}_1\text{H}^2 \rightarrow {}_2\text{He}^4 + Q$$
- $4(x_1 + x_2)$
 - $4(x_2 - x_1)$
 - $2(x_1 + x_2)$
 - $2(x_2 - x_1)$
16. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
- Assertion A:** The binding energy per nucleon decreases for a nucleus of large mass number.
- Reason R:** The coulomb repulsion between the protons makes the nucleus increasingly more stable.
- In the light of the above statements, choose the **correct** answer from the options given below:
- A is true but R is false.
 - A is false but R is true.
 - Both A and R are true and R is the correct explanation of A.
 - Both A and R are true but R is NOT the correct explanation of A.
17. The nuclei of which one of the following pairs of nuclei are isotones?
- ${}^{74}_{34}\text{Se}$, ${}^{71}_{31}\text{Ga}$
 - ${}^{92}_{42}\text{Mo}$, ${}^{92}_{40}\text{Zr}$
 - ${}^{84}_{38}\text{Sr}$, ${}^{86}_{38}\text{Sr}$
 - ${}^{40}_{20}\text{Ca}$, ${}^{32}_{16}\text{S}$
18. A nucleus of mass number A , originally at rest, emits an α -particle with speed v . The daughter nucleus recoils with a speed:
- $\frac{2v}{A+4}$
 - $\frac{4v}{A+4}$
 - $-\frac{4v}{A-4}$
 - $\frac{2v}{A-4}$
19. The Binding energy per nucleon of ${}^7_3\text{Li}$ and ${}^4_2\text{He}$ nuclei are 5.60 MeV and 7.06 MeV, respectively. In the nuclear reaction ${}^7_3\text{Li} + {}^1_1\text{H} \rightarrow {}^4_2\text{He} + {}^4_2\text{He} + Q$, the value of energy Q released is:
- 19.6 MeV
 - 2.4 MeV
 - 8.4 MeV
 - 17.3 MeV

20. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Nuclear binding energy (or mass defect) of hydrogen nucleus is zero.
Reason R: Hydrogen nucleus contain only one nucleon.
 In the light of the above statements, choose the **correct** answer from the options given below:
 (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is NOT the correct explanation of A.
21. If alpha particle and deuteron are moving with same velocity, the ratio of their respective de-Broglie wavelength will be:
 (1) 1 : 1
 (2) $1 : \sqrt{2}$
 (3) $\sqrt{3} : 1$
 (4) 1 : 2
22. What will be the ratio of de Broglie wavelengths of proton and α particle of same kinetic energy?
 (1) 2 : 1
 (2) 1 : 2
 (3) 4 : 1
 (4) 1 : 4
23. An electron jumps from n^{th} orbit to ground state in hydrogen atom, then the number of spectral lines emitted is:
 (1) $\frac{n-1}{2}$
 (2) $\frac{n(n-1)(n-2)}{2}$
 (3) $\frac{n(n-1)}{2}$
 (4) $\frac{n}{2}$
24. Hydrogen atom is excited from ground state to 7^{th} excited state. The number of lines in the emission spectrum will be:
 (1) 56 (2) 28
 (3) 30 (4) Infinite
25. The ratio of radius and velocity of electron in the n^{th} state of hydrogen atom is directly proportional to:
 (1) $1/n^3$ (2) n^3
 (3) n^2 (4) $1/n^2$
26. Magnetic field at the centre of hydrogen like atom due to motion of electron in the n^{th} orbit is proportional to [Z = atomic number]:
 (1) $\frac{Z^2}{n^2}$ (2) $\frac{Z^3}{n^2}$
 (3) $\frac{Z^3}{n^5}$ (4) $\frac{Z^2}{n^4}$
27. If an electron in a hydrogen atom jumps from the 3^{rd} orbit to the 2^{nd} orbit, it emits a photon of wavelength λ . When it jumps from the 4^{th} orbit to the 3^{rd} orbit, the corresponding wavelength of the photon will be:
 (1) $\frac{16}{25}\lambda$
 (2) $\frac{9}{16}\lambda$
 (3) $\frac{20}{7}\lambda$
 (4) $\frac{20}{13}\lambda$
28. An electron of a stationary hydrogen atom passes from the fifth energy level to the ground level. The velocity that the atom acquired as a result of photon emission will be [R = Rydberg constant, m = mass of hydrogen atom]:
 (1) $\frac{25m}{24hR}$
 (2) $\frac{24m}{25hR}$
 (3) $\frac{24hR}{25m}$
 (4) $\frac{25hR}{24m}$
29. The total energy of an electron in the first excited state of hydrogen is about -3.4eV . Its kinetic energy in this state is:
 (1) -3.4eV (2) -6.8eV
 (3) 6.8eV (4) 3.4eV
30. The angular momentum of an electron in a hydrogen atom is proportional to:
 (1) $\frac{1}{\sqrt{r}}$
 (2) $\frac{1}{r}$
 (3) $\sqrt{r}$
 (4) r^2

31. If an electron in hydrogen atom jumps from third orbit to second orbit, the frequency of the emitted radiation is given by (c is speed of light):

(1) $\frac{3Rc}{29}$ (2) $\frac{5Rc}{36}$
 (3) $\frac{7Rc}{36}$ (4) $\frac{8Rc}{31}$

32. Which of the following can't be the angular momentum of an electron orbiting in a hydrogen atom?

(1) $\frac{2h}{\pi}$
 (2) $\frac{h}{\pi}$
 (3) $\frac{3h}{2\pi}$
 (4) $\frac{5h}{4\pi}$

33. The ratio of shortest wavelengths of hydrogen atom in Lyman, Balmer and Paschen series are in the ratio of:

(1) 1 : 2 : 3
 (2) 1 : 4 : 9
 (3) 9 : 1 : 3
 (4) 4 : 12 : 17

34. The ratio of wavelengths of first line of Balmer series and that of the second line of Lyman series, in hydrogen spectrum is:

(1) $\frac{5}{27}$
 (2) $\frac{27}{5}$
 (3) $\frac{5}{32}$
 (4) $\frac{32}{5}$

35. The angular momentum of an electron in first orbit of Li^{++} ions is:

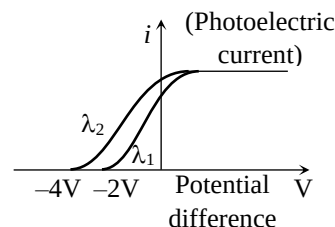
(1) $\frac{3h}{4\pi}$
 (2) $\frac{9h}{2\pi}$
 (3) $\frac{h}{2\pi}$
 (4) $\frac{h}{6\pi}$

SECTION - B

36. Outside a nucleus:

- (1) neutron is stable.
 (2) proton and neutron both are stable.
 (3) neutron is unstable.
 (4) neither neutron nor proton is stable.

37. The maximum value of stopping potential in the following diagram is:



- (1) -4V (2) -1V
 (3) -3V (4) -2V

38. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Photoelectric effect demonstrates the wave nature of light.

Reason R: The number of photoelectrons is proportional to the intensity of light.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is NOT the correct explanation of A.

39. A proton and an α -particle are accelerated through a potential difference of 100 V. The ratio of the wavelength associated with the proton to that associated with an α -particle is:

(1) $\sqrt{2}:1$
 (2) 2:1
 (3) $2\sqrt{2}:1$
 (4) $\frac{1}{2\sqrt{2}}:1$

40. If the kinetic energy of a free electron doubles, its de-Broglie wavelength changes by the factor:

(1) $\frac{1}{\sqrt{2}}$
 (2) $\sqrt{2}$
 (3) $\frac{1}{2}$
 (4) 2

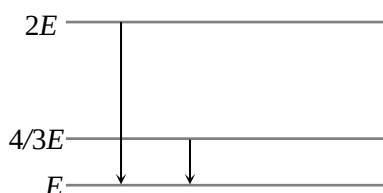
41. In the n^{th} orbit, the energy of an electron $E_n = -\frac{13.6}{n^2} \text{ eV}$ for hydrogen atom. The energy required to take the electron from first orbit to second orbit will be:

- (1) 10.2 eV
- (2) 12.1 eV
- (3) 13.6 eV
- (4) 3.4 eV

42. The Rutherford α -particle experiment shows that most of the α -particles pass through almost unscattered while some are scattered through large angles. What information does it give about the structure of the atom?

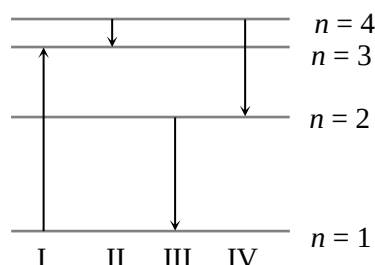
- (1) Most of the atom is hollow
- (2) The whole mass of the atom is concentrated in a small centre called nucleus
- (3) Nucleus is positively charged
- (4) All of these

43. The following diagram indicates the energy levels of a certain atom. When the system moves from $2E$ level to E , a photon of wavelength λ is emitted. The wavelength of photon produced during its transition from $\frac{4E}{3}$ level to E is:



- (1) $\lambda / 3$
- (2) $3\lambda / 4$
- (3) $4\lambda / 3$
- (4) 3λ

44. The diagram shows the energy levels for an electron in a certain atom. Which transition shown represents the emission of a photon with the most energy?



- (1) I
- (2) II
- (3) III
- (4) IV

45. Bohr's atom model assumes:

- (1) the nucleus is of infinite mass and is at rest.
- (2) electrons in a quantized orbit will not radiate energy.
- (3) mass of electron remains constant.
- (4) All of these

46. The ratio of the wavelengths for $2 \rightarrow 1$ transition in Li^{++} , He^+ and H is:

- (1) 1 : 2 : 3
- (2) 1 : 4 : 9
- (3) 4 : 9 : 36
- (4) 3 : 2 : 1

47. The radius of electron's second stationary orbit in Bohr's atom is R . The radius of the third orbit will be:

- (1) $3R$
- (2) $2.25R$
- (3) $9R$
- (4) $\frac{R}{3}$

48. The spectral series of the hydrogen spectrum that lies in the ultraviolet region is the:

- (1) Balmer series
- (2) Pfund series
- (3) Paschen series
- (4) Lyman series

49. When a hydrogen atom is raised from the ground state to an excited state?

- (1) P.E. increases and K.E. decreases
- (2) P.E. decreases and K.E. increases
- (3) Both kinetic energy and potential energy increase
- (4) Both K.E. and P.E. decrease

50. Nuclear fusion is possible:

- (1) only between light nuclei.
- (2) only between heavy nuclei.
- (3) between both light and heavy nuclei.
- (4) only between nuclei which are stable against β -decay.

SECTION-(II) CHEMISTRY**SECTION - A**

- 51.** Group 13 elements show:
- (1) only +1 oxidation state.
 - (2) only +3 oxidation state.
 - (3) +1 and +3 oxidation states.
 - (4) +1, +2 and +3 oxidation states.
- 52.** Number of electrons shared in the formation of nitrogen molecule is:
- (1) 6
 - (2) 10
 - (3) 2
 - (4) 8
- 53.** The first member of the p-block elements differs from the remaining members of their corresponding groups due to:
- (1) small size and absence of d-orbitals.
 - (2) diagonal relationship with other elements.
 - (3) difference in ability to form double and triple bonds.
 - (4) low ionisation enthalpy.
- 54.** Lassaigne's test (with silver nitrate) is commonly used to detect halogens such as chlorine, bromine and iodine but not useful to detect fluorine because the product AgF formed is:
- (1) volatile.
 - (2) reactive.
 - (3) explosive.
 - (4) soluble in water.
- 55.** Amongst group 13 elements, the least electropositive element is:
- (1) B
 - (2) Al
 - (3) In
 - (4) Ga
- 56.** The +1 oxidation state of thallium is more stable than its +3 oxidation state because of:
- (1) its atomic size.
 - (2) its ionisation potential.
 - (3) inert pair effect.
 - (4) diagonal relationship.
- 57.** Which of the following orders is **correct** regarding ionization enthalpy of group 14 elements?
- (1) $C > Si > Ge > Sn < Pb$
 - (2) $C \approx Si > Ge > Sn \approx Pb$
 - (3) $C < Si < Ge < Sn > Pb$
 - (4) $C < Si > Ge < Sn < Pb$
- 58.** The metallic character of the elements of group 14:
- (1) does not change.
 - (2) increases from top to bottom.
 - (3) decreases from top to bottom.
 - (4) has no significance.
- 59.** H_2S gas, on passing through an alkaline solution, forms white precipitate. The solution contains ions of:
- (1) Pb
 - (2) Zn
 - (3) Cu
 - (4) Ni
- 60.** Nitrogen forms N_2 , but phosphorus is converted into P_4 from P_2 the reason is:
- (1) Triple bond is present between phosphorus atom.
 - (2) $p\pi - p\pi$ bonding is strong.
 - (3) $p\pi - p\pi$ bonding is weak.
 - (4) Multiple bond is formed easily.
- 61.** Which of the following is the most stable oxidation state of aluminium?
- (1) -1
 - (2) +1
 - (3) +2
 - (4) +3
- 62.** Yellow ammonium sulphide solution is a suitable reagent used for the separation of:
- (1) HgS and PbS
 - (2) PbS and Bi_2S_3
 - (3) Bi_2S_3 and CuS
 - (4) CdS and As_2S_3
- 63.** Which of the following noble gases can form compounds with fluorine?
- (1) Helium (He)
 - (2) Neon (Ne)
 - (3) Xenon (Xe)
 - (4) All of these
- 64.** A metal chloride original solution (i.e. O.S) on mixing with K_2CrO_4 solution gives a yellow precipitate soluble in aqueous sodium hydroxide. The metal may be:
- (1) Mercury
 - (2) Iron
 - (3) Silver
 - (4) Lead
- 65.** The reason behind the lower atomic radius of Ga as compared to Al is:
- (1) poor screening effect of d-electrons for the outer electrons from increased nuclear charge.
 - (2) increased force of attraction of increased nuclear charge on electrons.
 - (3) increased ionisation enthalpy of Ga as compared to Al.
 - (4) anomalous behaviour of Ga.

66. Which of the following pair of anions are identified by conc. H_2SO_4 ?
- NO_3^- , CO_3^{2-}
 - Cl^- , NO_3^-
 - Br^- , CO_3^{2-}
 - CO_3^{2-}
67. Solution of a salt in dilute H_2SO_4 or acetic acid produces deep blue colour with starch iodide solution. The salt contains:
- Br^-
 - I^-
 - NO_3^-
 - NO_2^-
68. Ionization enthalpy ($\Delta_i H_1 \text{ kJ mol}^{-1}$) for the elements of group 13 follows the order.
- $\text{B} > \text{Al} > \text{Ga} > \text{In} > \text{Tl}$
 - $\text{B} < \text{Al} < \text{Ga} < \text{In} < \text{Tl}$
 - $\text{B} < \text{Al} > \text{Ga} < \text{In} > \text{Tl}$
 - $\text{B} > \text{Al} < \text{Ga} > \text{In} < \text{Tl}$
69. Among group 14 elements, which is a metalloid?
- C
 - Ge
 - Sn
 - Pb
70. Which of the following species has the highest dipole moment?
- NH_3
 - PH_3
 - AsH_3
 - SbH_3
71. Which of the following is a characteristic feature of the noble gases in Group 18 of the periodic table?
- High electronegativity
 - High ionization energy
 - High boiling points
 - They readily form covalent bonds with halogens
72. Which of the nitrates on strong heating leaves the metal as the residue?
- AgNO_3
 - $\text{Pb}(\text{NO}_3)_2$
 - $\text{Cu}(\text{NO}_3)_2$
 - $\text{Al}(\text{NO}_3)_3$
73. Which of the following is more stable?
- Ge^{4+}
 - Si^{4+}
 - Pb^{4+}
 - Sn^{4+}
74. Ba^{2+} ion is precipitated in fifth group as its:
- Oxides
 - Sulphates
 - Carbonates
 - Sulphides
75. Which is **true** regarding nitrogen?
- Less electronegative
 - Has low ionisation enthalpy
 - d-orbitals are available
 - Ability to form $p\pi-p\pi$ bonds with itself
76. The tendency of group 16 elements to show -2 oxidation state decreases as:
- $\text{S} > \text{Se} > \text{Te} > \text{Po}$
 - $\text{Po} > \text{Te} > \text{Se} > \text{S}$
 - $\text{Se} > \text{S} > \text{Po} > \text{Te}$
 - $\text{S} > \text{Se} > \text{Po} > \text{Te}$
77. Which noble gas is commonly used in neon signs?
- Xenon (Xe)
 - Krypton (Kr)
 - Neon (Ne)
 - Argon (Ar)
78. The ion that can **not** be precipitated by H_2S in presence of dil. HCl :
- Pb^{2+}
 - Bi^{3+}
 - Cu^{2+}
 - Co^{2+}
79. Among group 14 elements, the maximum number of catenated halides are formed by:
- carbon.
 - lead.
 - tin.
 - germanium.
80. Brown ring test is used to detect.
- Iodine
 - Nitrate
 - Iron
 - Bromide
81. Solution of ZnS in dil. HCl when treated with NaOH solution a white ppt. is formed which dissolves in excess of NaOH due to formation of:
- ZnCl_2
 - $\text{Zn}(\text{OH})\text{Cl}$
 - Na_2ZnO_2
 - None of these
82. What is the color of the precipitate formed when silver nitrate solution is added to a solution containing chloride ions?
- Yellow
 - White
 - Blue
 - Red
83. Which reagent is used to confirm the presence of sulphate ions in an unknown salt?
- Barium chloride solution
 - Silver nitrate solution
 - Sodium carbonate solution
 - Ammonium molybdate solution

84. The most stable +2 oxidation state is exhibited by:
 (1) Ge (2) Sn
 (3) Pb (4) Si
85. The stability of +1 oxidation state among Al, Ga, In and Tl increases in the sequence:
 (1) $Ga < In < Al < Tl$
 (2) $Al < Ga < In < Tl$
 (3) $Tl < In < Ga < Al$
 (4) $In < Tl < Ga < Al$

SECTION - B

86. In group 13 elements relative stability of +3 oxidation state may be given as:
 (1) $Tl^{3+} > In^{3+} > Ga^{3+} > Al^{3+} > B^{3+}$
 (2) $B^{3+} > Al^{3+} > Ga^{3+} > In^{3+} > Tl^{3+}$
 (3) $Al^{3+} > Ga^{3+} > Tl^{3+} > In^{3+} > B^{3+}$
 (4) $Al^{3+} > B^{3+} > Ga^{3+} > Tl^{3+} > In^{3+}$
87. Which of the following combines with Fe(II) ions to form a brown complex?
 (1) N_2O (2) NO
 (3) N_2O_3 (4) N_2O_4
88. Which of the following ions is confirmed by the formation of a white gelatinous precipitate when treated with ammonium hydroxide and ammonium chloride buffer solution?
 (1) Calcium (2) Magnesium
 (3) Aluminium (4) Iron(III)
89. The increasing order of atomic radii of the following group 13 elements is:
 (1) $Al < Ga < In < Tl$
 (2) $Ga < Al < In < Tl$
 (3) $Al < In < Ga < Tl$
 (4) $Al < Ga < Tl < In$
90. A mixture on treating with NaOH gives pungent smelling gas fuming with HCl. The mixture contains:
 (1) NH_4^+ (2) Na^+
 (3) Cd^{2+} (4) Cl^-
91. A brick red colour is imparted to Bunsen flame by a:
 (1) Ca salt (2) Sr salt
 (3) Na salt (4) Co salt
92. The property which is **not** true about fluorine is:
 (1) most of its reactions are exothermic.
 (2) it forms only one oxo acid.
 (3) highest electronegativity.
 (4) high F — F bond dissociation enthalpy.
93. What causes nitrogen (N_2) to be chemically inert?
 (1) High electronegativity of nitrogen
 (2) Absence of bond polarity
 (3) Short internuclear distance
 (4) High bond enthalpy
94. To avoid the precipitation of hydroxides of Ni^{2+} , Co^{2+} , Zn^{2+} and Mn^{2+} along with those of Fe^{3+} , Al^{3+} and Cr^{3+} the third group solution should be:
 (1) heated with a few drops of conc. HNO_3 .
 (2) treated with excess of NH_4Cl
 (3) concentrated
 (4) None of these
95. First ionization enthalpy of Al is lower than that of Mg. This is because:
 (1) the size of Al is bigger than Mg.
 (2) ionization enthalpy decrease in a period from left to right.
 (3) it is easier to remove electron from unpaired $3p^1$ than from paired $3s^2$
 (4) aluminium is a passive metal while magnesium is active.
96. Two test tubes containing a nitrate and a bromide are treated separately with H_2SO_4 . Brown fumes evolved are passed through water. The water will be coloured by vapours evolved from the test tube containing:
 (1) nitrate
 (2) bromide
 (3) both (1) and (2)
 (4) None of these
97. Which of the following halogen is liquid at room temperature?
 (1) Fluorine (2) Chlorine
 (3) Bromine (4) Iodine
98. The ionization energy of halogens increases as:
 (1) Fluorine < Chlorine < Bromine < Iodine
 (2) Chlorine < Fluorine < Bromine < Iodine
 (3) Iodine < Bromine < Chlorine < Fluorine
 (4) Iodine < Bromine < Fluorine < Chlorine
99. The **correct** order of increasing electron affinity of halogens is:
 (1) $I < Br < Cl$. (2) $Br < I < Cl$.
 (3) $Cl < Br < I$. (4) $I < Cl < Br$.
100. Nitrogen shows different oxidation states ranging from:
 (1) -3 to +5 (2) -5 to +5
 (3) 0 to -5 (4) -3 to +3

SECTION-(III) BOTANY

SECTION - A

101. The current species extinction rates are estimated to be _____ faster than in the pre-human times and our activities are responsible for the faster rates.

(1) 10 to 100 times
(2) 100 to 1,000 times
(3) 1 to 1,000 times
(4) 10 to 1,000,000 times

102. Pseudocopulation is an example of:

(1) wasp and fig.
(2) *Ophrys* and bees.
(3) *Ophrys* and wasp.
(4) bacteria and fungus.

103. Given below are two statements:

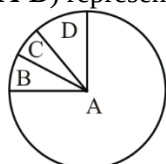
Statement I: The historic Convention on Biological Diversity ('The Earth Summit') was held in Rio de Janeiro.

Statement II: The World Summit on Sustainable Development was held in 2002 in Johannesburg, South Africa.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

104. Given below is the representation of the extent of global diversity of invertebrates. What groups the four portions (A-D) represent respectively?



(1)	Insects	Coelenterates	Arthropods	Molluscs
(2)	Annelids	Insects	Molluscs	Porifera
(3)	Insects	Molluscs	Crustaceans	Other animal groups
(4)	Molluscs	Arthropods	Coelenterates	Insects

105. How many of the following are **correct** for sacred groves and their location?

A. Aravalli Hills - Rajasthan
B. Western Ghat regions - Karnataka
C. Chanda and Bastar areas - Madhya Pradesh
D. Sarguja – Gujrat

Choose the most appropriate answer from the options given below:

(1) One
(2) Two
(3) Three
(4) Four

106. Which of the following is **not** a biodiversity hotspot?

(1) Western Ghats
(2) Sri Lanka
(3) Indo-Burma
(4) Madhya Pradesh

107. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Endemism	(I)	Khasi and Jaintia hills
(B)	Hotspot of India	(II)	Species found in a particular area only
(C)	Sacred grove	(III)	Advanced <i>ex-situ</i> Conservation
(D)	Cryopreservation	(IV)	Western Ghats

Choose the **correct** answer from the options given below:

(1) A-II, B-III, C-IV, D-I
(2) A-III, B-II, C-I, D-IV
(3) A-IV, B-III, C-II, D-I
(4) A-II, B-IV, C-I, D-III

108. The 'Evil Quartet' of biodiversity loss includes all **except**:

(1) habitat loss.
(2) introduction of alien species.
(3) over-exploitation.
(4) hunting.

109. The slope of the line is much steeper when the species-area relationships among _____ areas are considered.

(1) large
(2) very large
(3) less
(4) very small

110. Rivets on the wings represent:

(1) exotic species.
(2) endemic species.
(3) key species.
(4) indigenous species.

111. Read the following statement/s and mark the **right** option.

- A. The accelerated rates of species extinctions that the world is facing now are largely due to human activities.
- B. The most dramatic examples of habitat loss come from tropical rain forests.
- C. Ecologists warn that if the present trends continue, nearly half of all the species on earth might be wiped out within the next 100 years.
- D. Amazon rain forest is known for having the highest biodiversity in the world.

Choose the most appropriate answer from the options given below:

- (1) C is correct only
- (2) B and C are correct only
- (3) A and C are correct only
- (4) All A, B, C and D are correct

112. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: In India, ecologically unique and biodiversity-rich regions are legally protected as biosphere reserves, national parks and sanctuaries.

Reason R: India has a history of religious and cultural traditions that emphasised protection of nature.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

113. Read the following and select the **correct** option.

- (1) In terrestrial ecosystem smaller fraction of energy flow occurs through detritus food chain(DFC).
- (2) Artificial interconnection of food chain makes it a food web.
- (3) DFC is not at all connected with grazing food chain at any level.
- (4) Decomposers are mainly fungi and bacteria.

114. Which one of the following animals may occupy more than one trophic levels in the same ecosystem at the same time?

- (1) Sparrow (2) Lion
- (3) Goat (4) Frog

115. Each trophic level has a certain mass of living material at a particular time called as the:

- (1) standing state. (2) standing life.
- (3) standing crop. (4) productivity.

116. If in a pond, there were 20 lotus last year and through reproduction 8 new plants are added, taking current population to 28, the birth rate per year is?

- (1) 0.2 (2) 0.4
- (3) 0.6 (4) 0.8

117. _____ is the number of individuals of the population who left the habitat and gone elsewhere during the time period under consideration.

- (1) Immigration
- (2) Emigration
- (3) Natality
- (4) Mortality

118. Match **List-I** with **List-II**.

List-I		List-II	
(A)	<i>Cuscuta</i> and Mango tree	(I)	Commensalism
(B)	Cuckoo and crow	(II)	Brood parasitism
(C)	Sea anemone and clown fish	(III)	Parasitism
(D)	Female wasp and fig	(IV)	Mutualism

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
- (2) A-I, B-II, C-III, D-IV
- (3) A-IV, B-II, C-I, D-III
- (4) A-III, B-II, C-I, D-IV

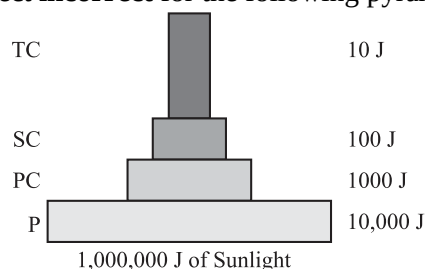
119. Which of the following statements related to stratification are **incorrect**?

- A. Horizontal distribution of different species.
- B. Vertical distribution of different species.
- C. Species occupy different levels.
- D. Species occupy same levels.

Choose the most appropriate answer from the options given below:

- (1) A and C (2) A and D
- (3) A and B (4) B and C

120. Select **incorrect** for the following pyramid.



- (1) It represents pyramid of energy.
- (2) It is upright.
- (3) It follows 10% energy law transfer.
- (4) It can be inverted in case of aquatic ecosystem.

121. There is no difference between:
- (1) secondary consumers and herbivores.
 - (2) primary consumers and herbivores.
 - (3) first trophic level and herbivores.
 - (4) primary carnivores and second trophic level.
122. From the given set of the ecosystem, how many are included in terrestrial ecosystem?
- Wetland, estuary, forest, desert, river, grassland, lake and pond
- Choose the most appropriate answer from the options given below:
- (1) Two
 - (2) Three
 - (3) Four
 - (4) Five
123. Read the following statements and mark the **correct** option.
- A. Gross primary productivity is always greater than the net primary productivity.
 - B. Primary productivity depends on the plant species inhabiting a particular area.
 - C. Productivity also depends on a variety of environmental factors, availability of nutrients and photosynthetic capacity of plants.
 - D. A constant input of solar energy is the basic requirement for any ecosystem to function and sustain.
- Choose the most appropriate answer from the options given below:
- (1) C is correct only
 - (2) B and C are incorrect only
 - (3) A and C are incorrect only
 - (4) All A, B, C and D are correct
124. Which one of the following statements is **correct** about decomposition?
- (1) Decomposition rate is slower if detritus is rich in lignin and chitin, and quicker if detritus is rich in nitrogen and water - soluble substances like sugars.
 - (2) Decomposition rate is slower if detritus is rich in nitrogen and water - soluble substances like sugars, and quicker if detritus is rich in lignin and chitin.
 - (3) Decomposition rate is slower if detritus is rich in cellulose, and quicker if detritus is rich in phosphorus.
 - (4) Decomposition rate is quicker if detritus is rich in lignin, and lower if detritus is rich in sulphur.
125. Given below are two statements:
- Statement I:** India has a greater ecosystem diversity than Greenland.
- Statement II:** According to the IUCN (2004), the total number of plant and animal species described so far is slightly more than 15 million.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
 - (2) Statement I is incorrect but Statement II is correct.
 - (3) Both Statement I and Statement II are correct.
 - (4) Both Statement I and Statement II are incorrect.
126. Select the **incorrect** statement.
- (1) Food chains or rather webs are formed such that an animal feeds on a plant or on another animal and in turn is food for another.
 - (2) The chain or web is formed because of this interdependency.
 - (3) The energy trapped by the producer, hence, is either passed on to a consumer or the organism dies.
 - (4) The energy that is trapped into an organism remains in it forever.
127. Ecology is basically concerned with which level/s of biological organisation.
- A. Organisms.
 - B. Biomes.
 - C. Communities.
 - D. Populations.
- Choose the most appropriate answer from the options given below:
- (1) C is correct only
 - (2) B and C are correct only
 - (3) A and C are correct only
 - (4) All A, B, C and D are correct
128. Gause's 'Competitive Exclusion Principle' states that:
- (1) humans are the most widespread agents of disturbance.
 - (2) in a competition for similar resource both the participants are benefitted.
 - (3) in a competition, both the participants are excluded.
 - (4) two closely related species competing for the same resources cannot co-exist indefinitely and competitively inferior one will be eliminated eventually.
129. When all the starfish were removed from an enclosed intertidal area, more than _____ species of invertebrates became extinct within a year, because of interspecific competition.
- (1) 10
 - (2) 20
 - (3) 30
 - (4) 50

130. Lichens represent an intimate ____ (i) ____ relationship between a fungus and ____ (ii) ____ . Identify (i) and (ii) respectively.
- (1) (i)-parasitic; (ii) - bacteria
 - (2) (i)-mutualistic; (ii) - cyanobacteria
 - (3) (i)-predation; (ii) - cyanobacteria
 - (4) (i)-mutualistic; (ii) - bacteria

131. Select the **correct** pair.
- (1) Bee and orchid – competition
 - (2) Liver fluke – Ectoparasite
 - (3) Barnacle and whale– Amensalism
 - (4) Mycorrhizae – Mutualism

132. $\frac{dN}{dt} = rN \left(\frac{K - N}{K} \right)$ represents which type of growth?
- (1) *Verhulst-Pearl* logistic growth.
 - (2) *Connell's* logistic growth.
 - (3) Maxwells exponential growth.
 - (4) *Verhulst-Pearl* exponential growth.

133. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Predators in nature are 'prudent'.

Reason R: If a predator is too efficient and overexploits its prey, then the prey might become extinct and following it, the predator will also become extinct for lack of food.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

134. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The size of a population for any species is not a static parameter.

Reason R: It keeps changing with time, depending on various factors including food availability, predation pressure and adverse weather.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

135. In growth pattern, $(1 - N/K)$ is:
- (1) carrying capacity.
 - (2) intrinsic rate of natural increase.
 - (3) environmental resistance.
 - (4) biotic potential.

SECTION - B

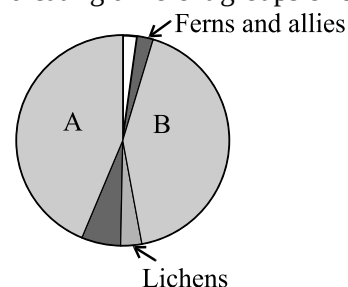
136. Given below are two statements:

Statement I: Many ecologists regard the entire biosphere as a global ecosystem, as a composite of all local ecosystems on Earth.

Statement II: Crop fields and an aquarium may also be considered as man-made ecosystems, In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

137. Select **correct** for **A** and **B** from the following pie chart indicating different groups of organisms.



- (1) A-Algae and B-Bryophytes
- (2) A-Fungi and B-Angiosperms
- (3) A-Fungi and B-Bryophytes
- (4) A-Bryophytes and B-Algae

138. How many of the following is/are **correct** regarding predator?

- A. It keeps prey population under control.
- B. It helps in maintaining species diversity.
- C. Biological control methods adopted in agricultural pest control are based on the ability of the predator to regulate prey population.
- D. Predation is nature's way of transferring to higher trophic levels the energy fixed by plants.

Choose the most appropriate answer from the options given below:

- (1) One
- (2) Two
- (3) Three
- (4) Four

139. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Frog	(I)	Chemical defence
(B)	Prickly pear cactus	(II)	Moth
(C)	Opium	(III)	Morphological defence
(D)	Thorns	(IV)	Camouflage

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
- (2) A-III, B-II, C-I, D-IV
- (3) A-IV, B-II, C-I, D-III
- (4) A-II, B-I, C-IV, D-III

140. Energy storage at consumer level is called:

- (1) gross primary productivity.
- (2) secondary productivity.
- (3) net primary productivity.
- (4) net productivity.

141. Productivity is expressed in terms of:

- (1) $\text{gm}^{-2} \text{yr}^{-1}$.
- (2) $\text{kcal m}^{-1} \text{yr}^{-1}$.
- (3) $\text{kcal m}^2 \text{yr}^{-1}$.
- (4) $\text{gm}^2 \text{yr}^{-1}$.

142. If N is the population density at time t , then its density at time $t + 1$ is:

- (1) $N_{t+1} = N_t + [(B + D) - (I + E)]$.
- (2) $N_{t+1} = N_t + [(B + E) - (D + I)]$.
- (3) $N_{t+1} = N_t + [(B + I) - (D + E)]$.
- (4) $N_{t+1} = N_t + [(B + I) + (D - E)]$.

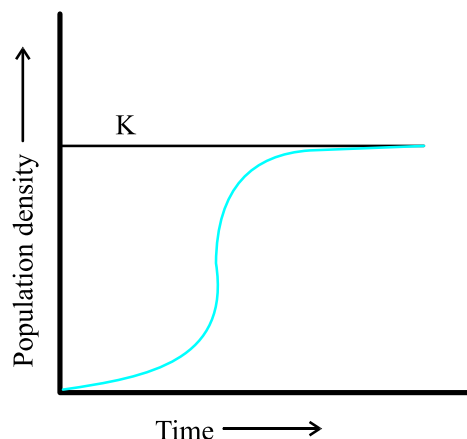
143. The cattle egret and grazing cattle show:

- (1) symbiosis.
- (2) amensalism.
- (3) ectoparasitism.
- (4) commensalism.

144. What is the phenomenon called in which two species competing for the same resources can avoid competition by choosing?

- (1) Resource structuring
- (2) Competition
- (3) Predation
- (4) Resource partitioning

145. What does the blue line in the given figure indicate?



- (1) Decreasing growth
- (2) Sigmoidal growth
- (3) Exponential growth
- (4) J-shaped

146. The total number of national parks and sanctuaries in India are:

- (1) 82 national parks and 4101 sanctuaries.
- (2) 83 national parks and 421 sanctuaries.
- (3) 85 national parks and 450 sanctuaries.
- (4) 90 national parks and 448 sanctuaries.

147. Which of the following related to diversity and number of species are **correct**?

- A. There are more than 20,000 species of ants.
- B. There are more than 4,00,000 species of beetles.
- C. There are more than 28,000 species of fishes.
- D. Nearly 20,000 species of orchids.

Choose the most appropriate answer from the options given below:

- (1) A and B only
- (2) A, C and D only
- (3) A and D only
- (4) A, B, C and D

148. Energy captured by plants to perform photosynthesis is approximately:

- (1) 2-10% of incident solar radiation.
- (2) 2-10% of photosynthetically active radiation.
- (3) 50% of incident solar radiation.
- (4) 50% of photosynthetically active radiation.

149. Select the **mismatched** pair w.r.t. extinct animals and region.

- (1) Dodo - Mauritius
- (2) Quagga - Africa
- (3) Stellar's Sea Cow - India
- (4) Thylacine - Australia

150. The species-area relationship on log scale is generally a:

- (1) rectangular hyperbolic function.
- (2) line function.
- (3) sine function.
- (4) curve function.

SECTION-(IV) ZOOLOGY

SECTION - A

151. Select the **correct** option.

- (1) *Dryopithecus* was more man like.
- (2) *Homo habilis* probably did not eat meat.
- (3) *Homo erectus* had a cranial capacity of 700 cc.
- (4) Agriculture came around 18000 years ago.

152. Match **List-I** with **List-II**.

List-I		List-II	
(A)	<i>Taq</i> polymerase	(I)	Cleaves the ends of linear DNA.
(B)	Exonuclease	(II)	Breakdown of fungal cell wall.
(C)	Protease	(III)	Stable above 90°C
(D)	Chitinase	(IV)	Degradation of protein

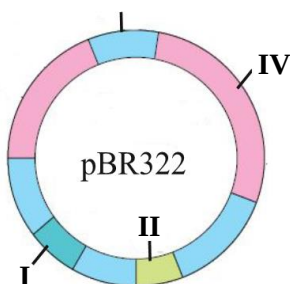
Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
- (2) A-IV, B-III, C-I, D-II
- (3) A-II, B-I, C-IV, D-III
- (4) A-III, B-I, C-IV, D-II

153. Which of the following diseases has a transgenic model for research purposes?

- (1) Diabetes
- (2) Hypertension
- (3) Alzheimer's
- (4) Common cold

154. Following is the diagrammatic representation of *E. coli* vector pBR322. Which of the given options **correctly** identifies its components?



- (1) I - rop
- (2) II - amp^r
- (3) III - ori
- (4) IV - tet^r

155. Given below are two statements:

Statement I: 'Cry' proteins are named so because they are crystal proteins.

Statement II: Transgenic mice are used to test vaccine safety.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

156. Choose the **incorrect** statement.

- (1) Alfred Wallace worked in Malay Archipelago.
- (2) Vertebrate hearts and brains exhibit convergent evolution.
- (3) Branching descent and natural selection are two key concepts of Darwinian theory.
- (4) Similar habitat results in selection of similar adaptive features in different organisms.

157. Read the following statements (**A-D**).

- A. Restriction enzymes belong to a large class of enzymes called nuclease.
- B. *Hind II* cuts DNA at a specific sequence of six base pairs.
- C. Each restriction endonuclease recognises a specific palindromic nucleotide sequence in the DNA.
- D. Restriction enzymes cut the sugar-phosphate backbone of DNA.

Which of the above given statements is/are **correct** for restriction enzymes?

- (1) A, B and D only
- (2) A and D only
- (3) A, C and D only
- (4) A, B, C and D

158. Why is traditional knowledge related to bio-resources important for developing countries?

- (1) It is primarily beneficial for industrialised nations.
- (2) It can be exploited to develop modern applications and save costs in commercialisation.
- (3) It is of no value in scientific research.
- (4) It leads to decreased biodiversity.

159. Choose the **incorrect** match.

Placental Mammals		Australian marsupials
(1)	Flying squirrel	Flying phalanger
(2)	Anteater	Numbat
(3)	Koala	Bobcat
(4)	Wolf	Tasmanian wolf

160. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The recombinant DNA molecule cannot be created, unless one cuts the vector and source DNA with same restriction enzyme.

Reason R: When cut by same enzyme both will have same sticky ends, that can be easily joined by DNA ligase.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

161. What is main purpose of creating transgenic animals?

- (1) To enhance their physical appearance.
- (2) To study how genes are regulated and affect body functions.
- (3) To increase their lifespan.
- (4) To improve their reproductive rates.

162. Insertion of gene of interest in selectable marker leads to _____ of selectable marker.

Choose the option which fills the blank **correctly**.

- (1) insertional activation
- (2) insertional inactivation
- (3) duplication
- (4) hyperactivation

163. Which of the following theories suggests that species existed in their current forms since creation?

- (1) Natural selection.
- (2) Panspermia.
- (3) Special creation.
- (4) Abiogenesis.

164. Which of the following was the first to be used as biopesticide on the commercial scale in the world?

- (1) *Bacillus thuringiensis*
- (2) *Escherichia coli*
- (3) *Pseudomonas aeruginosa*
- (4) *Agrobacterium tumefaciens*

165. Given below are two statements:

Statement I: RNA interference is a natural process in eukaryotes used to silence specific genes.

Statement II: Insulin was originally extracted from pancreas of slaughtered pigs and cattle.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

166. What was the name of the sail ship used by Charles Darwin during his sea voyage?

- (1) HMS Beagle
- (2) HSM Beagle
- (3) HMS Eagle
- (4) HSM Eagle

167. If a recombinant DNA bearing gene for ampicillin resistance is transferred successfully intact into *E. coli* cells and the host cells, are spread on agar plates containing ampicillin, then:

- (1) both transformed and non-transformed recipient cells will die.
- (2) both transformed and non-transformed recipient cell will grow.
- (3) transformed recipient cells will grow and non-transformed recipient cells will die.
- (4) transformed recipient cells will die and non-transformed recipient cells will grow.

168. The **correct** chronological order of human evolution from earliest to recent is:

- (1) *Australopithecines* → *Ramapithecus* → *Homo habilis* → *Homo erectus*
- (2) *Ramapithecus* → *Australopithecines* → *Homo habilis* → *Homo erectus*
- (3) *Ramapithecus* → *Homo habilis* → *Australopithecines* → *Homo erectus*
- (4) *Australopithecines* → *Homo habilis* → *Ramapithecus* → *Homo erectus*

- 169.** What is the primary purpose of the Genetic Engineering Approval Committee (GEAC) in India?
- (1) To grant patents for genetically modified organisms.
 - (2) To regulate genetic modification research and assess the safety of GM organisms.
 - (3) To promote the use of traditional farming methods.
 - (4) To ban all forms of genetic engineering.

170. Read the following statements (A-E).

- A. If any protein-encoding gene is expressed in a heterologous host, it is called a recombinant protein.
- B. Bacterial cells are made 'competent' to take up DNA by treating them with a specific concentration of a divalent cation.
- C. Recombinant DNA can be forced into cells by placing the cells with recombinant DNA in warm water.
- D. Disarmed pathogen vectors can be used to transfer the recombinant DNA into the host.
- E. The RNA can be removed by treatment with ribonuclease.

Which of the following statements are **correct**?

- (1) A, C and D only
- (2) A, B, D and E only
- (3) A and D only
- (4) A, B, C, D and E

171. An American company prepared two DNA sequences corresponding to A and B chains of human insulin and introduced them through a plasmid into:

- (1) *Entamoeba histolytica*.
- (2) *Escherichia coli*.
- (3) *Salmonella typhimurium*.
- (4) *Haemophilus influenzae*.

172. Read the following statements (A-D) w.r.t. genetic modification in plants.

- A. It has reduced reliance on chemical pesticides.
- B. It increased post-harvest losses.
- C. It has increased efficiency of minerals usage by the plants.
- D. It has made crops more sensitive to cold, heat and drought.

Which of the statements given above are **correct**?

- (1) A and C only
- (2) A, B and C only
- (3) B, C and D only
- (4) C and D only

173. What is the primary goal of inserting alien DNA into a cloning vector and transferring it into a host cell?

- (1) To enhance the genetic diversity of the host organism.
- (2) To produce a small number of copies of the alien DNA.
- (3) To express and produce a desirable protein.
- (4) To disrupt the host cell's DNA.

174. What issue does the second amendment of the Indian Patents Bill address?

- (1) Decreasing agricultural output
- (2) Unauthorized exploitation of bio-resources
- (3) Promotion of foreign patents
- (4) Ban on traditional knowledge

175. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Methylation	(I)	Extraction of DNA from the gel plate
(B)	1963	(II)	Virus that infects bacteria.
(C)	Bacteriophage	(III)	Addition of methyl group to DNA
(D)	Elution	(IV)	Year when the enzymes restricting bacteriophage growth were discovered.

Choose the **correct** answer from the options given below:

- (1) A-IV, B-II, C-III, D-I
- (2) A-III, B-I, C-IV, D-II
- (3) A-I, B-III, C-IV, D-II
- (4) A-III, B-IV, C-II, D-I

176. Which of the following best describes the relationship between geological and biological history?

- (1) They are unrelated.
- (2) Geological history of earth closely correlates with its biological history.
- (3) Biological history is older than geological history.
- (4) They are both static and unchanging.

177. What major event is described by the Big Bang theory?

- (1) The continuous expansion of the earth.
- (2) The formation of the planets.
- (3) A singular explosive event marking the universe's origin.
- (4) The cooling of Earth's surface.

178. Identify the **correct** statement w.r.t electrophoresis.

- (1) DNA is loaded on gel at the anode terminal.
- (2) Gel concentration does not affect movement of DNA.
- (3) Smaller the size of DNA fragment large is the distance it travels through it.
- (4) Pure DNA can be visualised directly by exposing UV radiation.

179. What is an antisense technology?

- (1) A cell displaying a foreign antigen used for synthesis of antigens.
- (2) Production of soma clonal variants in tissue in tissue cultures.
- (3) When a piece of RNA that is complementary in sequence is used to stop expression of a specific gene.
- (4) RNA polymerase produces DNA.

180. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Dark winged moth survived after industrialisation.

Reason R: The tree trunks became dark due to industrial smoke and soot due to which dark winged moth easily camouflaged.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

181. What was the outcome of Pasteur's experiments regarding spontaneous generation?

- (1) It proved that life arises from non-living matter.
- (2) It confirmed the theory of spontaneous generation.
- (3) It disproved the theory of spontaneous generation.
- (4) It provided evidence for panspermia.

182. The main challenge for production of insulin using rDNA techniques was:

- (1) splitting A and B- peptide chains.
- (2) addition of C-peptide to proinsulin.
- (3) getting insulin assembled to mature form.
- (4) removal of C- peptide from active insulin.

183. What ethical issue arises from genetic modification of organisms?

- (1) Increased biodiversity
- (2) Potential unpredictable effects on ecosystems
- (3) Enhanced agricultural productivity
- (4) Improved health benefits

184. Given below are two statements:

Statement I: Sexual reproduction preserves the genetic information, while asexual reproduction permits variation.

Statement II: Each restriction endonuclease functions by 'inspecting' the length of a DNA sequence.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

185. For an effective treatment of a disease:

- (1) early diagnosis is needed but understanding of its pathophysiology is not needed.
- (2) early diagnosis is not needed but understanding of its pathophysiology is needed.
- (3) early diagnosis and understanding of its pathophysiology are needed.
- (4) neither early diagnosis nor understanding of its pathophysiology is needed.

SECTION - B

186. X coloured bands of DNA can be visualised if an ethidium bromide-stained gel exposed to Y light.

Choose the option which fills X and Y **correctly**.

- | X | Y |
|------------|---------|
| (1) Red | visible |
| (2) Grey | UV |
| (3) Orange | UV |
| (4) Purple | visible |

187. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: ADA deficiency disorder is caused due to the excessive synthesis of gene for adenosine deaminase.

Reason R: Deficiency of ADA affects the human immune system.

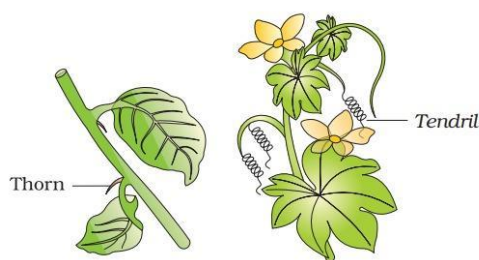
In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

188. In *Bt* cotton, the *Bt* toxin present in plant tissue as pro-toxin is converted into active toxin due to:

- (1) Alkaline pH of the insect gut.
- (2) Acidic pH of the insect gut.
- (3) Action of gut microorganisms.
- (4) Presence of conversion factors in insect gut.

189. The following given diagrams show;



- (1) convergent evolution.
- (2) divergent evolution.
- (3) natural selection.
- (4) anthropogenic action.

190. Given below are two statements:

Statement I: Bacteriophages have low copy numbers in the bacterial cells.

Statement II: The first step of genetic modification is identifying DNA with useful genes. In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

191. Which of the following initiatives is part of the Indian Patents Bill amendment?

- (1) Giving inadequate compensations.
- (2) Research and development initiatives.
- (3) Promote unauthorised exploitation of bio-resources.
- (4) Giving unequal benefit sharing.

192. Read the following statements (A-D).

A. The brain capacities of *Homo habilis* were between 650-800cc.

B. Darwin's finches are small black birds.

C. *Ichthyosaurs* was fish like reptile.

D. The original finches were seed-eating.

Which of the above statements are **correct**?

- (1) A, B and C only
- (2) A and B only
- (3) A, B and D only
- (4) A, B, C and D

193. Polymerase chain reaction is used in:

- (1) confirming presence of a pathogen during early infection.
- (2) identifying the mutated genes in suspected cancer patients.
- (3) detection of the presence of HIV in suspected AIDS patient.
- (4) all of these

194. Match List-I with List-II.

List-I		List-II	
(A)	Ti-plasmid	(I)	amp ^R
(B)	PCR	(II)	Join the sticky ends of DNA fragments
(C)	Ligase	(III)	<i>Agrobacterium</i>
(D)	<i>PvuI</i>	(IV)	Amplification of genetic material

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-IV, D-III
- (2) A-III, B-IV, C-II, D-I
- (3) A-I, B-II, C-III, D-IV
- (4) A-I, B-IV, C-II, D-III

195. Which of the following shows embryological evidence of evolution?

- (1) Fossils in different sedimentary layers.
- (2) Vestigial gill slits in embryos of all vertebrates.
- (3) Forelimbs of mammals.
- (4) Potato and sweet potato.

196. In the presence of chromogenic substrate, recombinant bacteria give:
- (1) colourless colonies.
 - (2) blue colonies.
 - (3) black colonies.
 - (4) green colonies.
197. A permanent cure for ADA deficiency could be:
- (1) Enzyme replacement therapy.
 - (2) Bone marrow transplant.
 - (3) Introduction of a functional ADA cDNA into lymphocytes.
 - (4) Introduction of gene isolated from marrow cells producing ADA into early embryonic cells.
198. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The tumour inducing plasmid (Ti plasmid) acts as cloning vector in recombinant DNA technology.
Reason R: The Ti plasmid which is used in mechanisms of delivering genes to a cell remains pathogenic.
In the light of the above statements, choose the **correct** answer from the options given below:
- (1) A is true but R is false.
 - (2) A is false but R is true.
 - (3) Both A and R are true and R is the correct explanation of A.
 - (4) Both A and R are true but R is not the correct explanation of A.

199. Which of the following statements is **true** regarding Darwin's theory of evolution?
- (1) It implies all species are static and unchanging.
 - (2) It suggests that environmental factors influence survival.
 - (3) It asserts that species have no common ancestors.
 - (4) It denies the occurrence of extinction.
200. Read the following statements (A-E).
- A. *In vitro* fertilisation leading to a 'test-tube' baby technique is included in biotechnology.
 - B. Retroviruses in animals have the ability to transform normal cell into cancerous cells.
 - C. To recover many copies of target DNA, it should be cloned in a vector with a low-copy-number origin.
 - D. Mosquito acts as an insect vector.
 - E. Sterile ambience in chemical engineering processes enable growth of only the desired microbe.
- Which of the following statements is/are **correct**?
- (1) A and B only
 - (2) A, B, C and D only
 - (3) A, B, D and E only
 - (4) A, B, C, D and E

